



Project Title

Product Search and Recommendation System for Shopping

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

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Abstract

The **Product Search and Recommendation System for Shopping** is designed to make online product searching faster, easier, and more accurate. When shopping websites contain thousands of products, users may have difficulty finding the right product because of spelling mistakes, incomplete product names, or unclear search requirements. Our system provides a smart product search facility where users can search using product names, keywords, or descriptive queries. It can correct common spelling mistakes, provide auto-complete suggestions, and allow users to filter products based on price, brand, and category. This helps users quickly reach the products that match their requirements instead of going through a large number of unrelated results.

A key feature of the system is intent-based search, where the system identifies the user's actual requirement from a search query. For example, when a user searches for “laptop for gaming under \$1000,” the system identifies the product category, purpose, and budget, and uses these requirements to find and rank suitable products. The system also recommends relevant products and provides a simple explanation of why each product was recommended, such as matching the user's budget, category, features, or preferences. Thus, the project combines product search, spelling correction, auto-complete, filtering, intent understanding, ranking, and explainable recommendations to provide a more convenient and informed shopping experience.